

EV Pulse NC

Prioritizing North Carolina's \$109M EV Charging Investment

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Economics · Spring 2026



Four Acts

The full paper runs ~108 pages. This talk is the spine.



1

Act 1

The Problem

2

Act 2

How I Did It

3

Act 3

What I Found

4

Act 4

What It Means

The \$109M NEVI Question

The Opportunity

Federal NEVI Formula Program: **\$109M to North Carolina**, \$5B nationally — the largest single federal investment in EV infrastructure in U.S. history.

The Missing Piece

No validated, publicly available method exists for **county-level prioritization**. The default outcome reinforces existing concentration in the Research Triangle and Charlotte metro — efficiency vs. equity, unresolved.



This is a live federal allocation decision with no defensible answer yet on the table.



ACT 1 — THE PROBLEM

Explosive Growth — Infrastructure Has Not Kept Pace

1,727%

BEV Fleet Growth

Sep 2018 → Jun 2025

53.8%

CAGR

Compound annual rate over 7 years

73%

Fleet Concentration

Top 10 counties hold 73% of statewide BEVs

4.34%

Forecast MAPE

Validated on 400-observation holdout

82 months of NCDOT panel data · 8,200+ county-month observations · Forecast validated on a true out-of-sample holdout. The demand side is clear. The supply-side question is what this study answers.

Five Research Questions

Five questions structure the work — each maps directly to one of the five analytical phases.

01

Forecast Accuracy

How accurate are the county-level BEV adoption forecasts?

02

Demand-Supply Gaps

Where are the largest gaps at the county level?

03

Sub-County Inequality

How do ZIP-level density patterns reveal priorities that aggregation hides?

04

Commuter Flows

How do workplace commuting patterns reshape infrastructure demand?

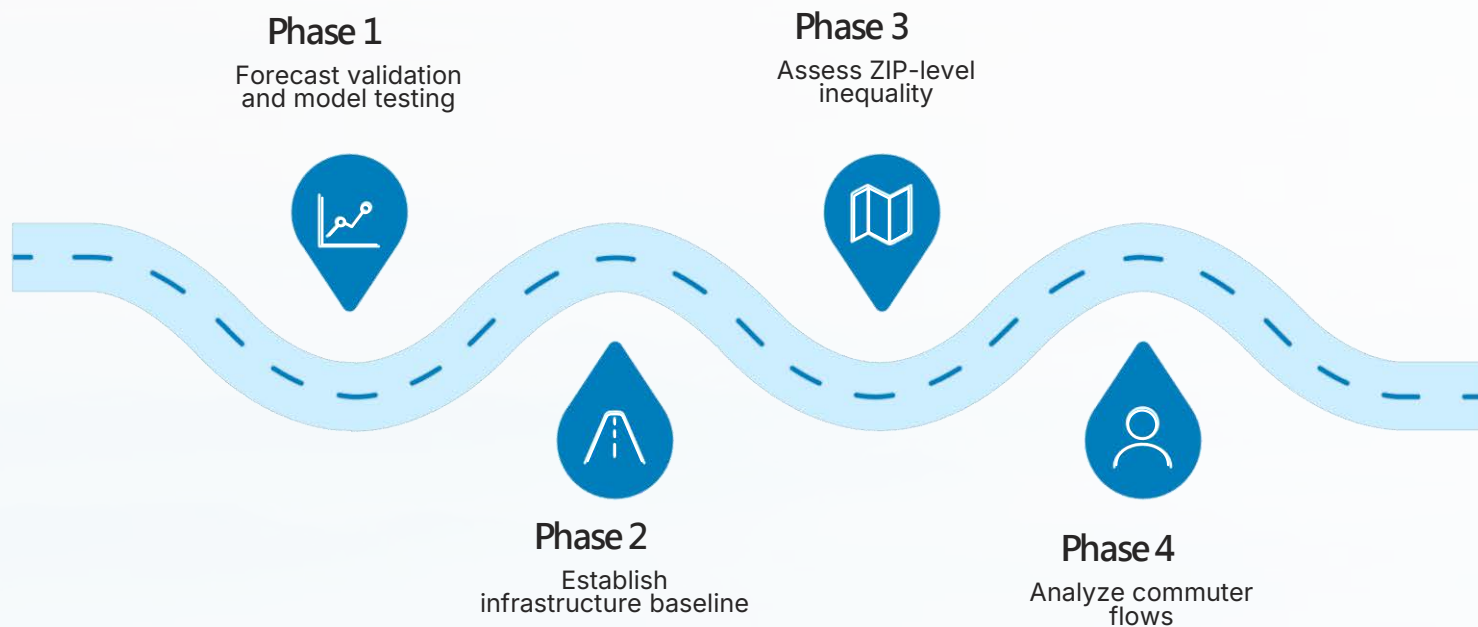
05

Defensible Rankings

Can a weighted scoring equation produce defensible NEVI funding recommendations?

Five-Phase Analytical Pipeline

Each phase produces committed code, processed data, and validation evidence: the discipline that makes a sole-author capstone tractable.



Phase 1

Forecast validation — SAS Model Studio out-of-sample testing

Phase 2

Infrastructure baseline — AFDC API: 1,985 stations, 6,145 connectors

Phase 3

ZIP-level inequality — Gini + additive Theil-T dec

Phase 4

Workplace charging demand — LEHD LODS 2021 commuter flows

Phase 5

Justice40 equity overlay — CEJST v2.0 + area-weighted crosswalk

Data Sources: All Public, All Reproducible

Nothing in this analysis depends on private or licensed data. A federal reviewer can re-execute the entire pipeline from a public GitHub repository.

Source	Content	Coverage
NCDOT	Monthly BEV registration counts	Sep 2018 – Oct 2025
NREL AFDC API	Charging-station inventory	Feb 2026 pull
U.S. Census	ACS demographics + TIGER geographies	2021 5-year ACS
LEHD LODS 2021	Origin-destination commuter flows	2021 annual
CEJST v2.0	Justice40 disadvantaged tract designations	Archived via EDGI/PEDP

✔ Every source is federal or state-public. The pipeline is audit-ready.

Methodology Highlights

Six techniques carry the analytical weight — each addressing a specific defensibility lane.



Forecast Validation

Out-of-sample MAPE + formal Chow structural-break testing at IRA passage



VIF Diagnostics

Scoring-pillar orthogonality confirmed — max VIF 1.41, well below the 5.0 threshold



Theil-T Decomposition

Between/within partitioning of the GE(1) class — verified to machine precision (2.22e-16)



Area-Weighted Crosswalk

Tract-to-ZCTA interpolation meeting EPA EJScreen + HUD USPS standards



23/23 Validation Suite

All 23 federal-standard validation checks passed across all methodology lanes



AI-Assisted Review

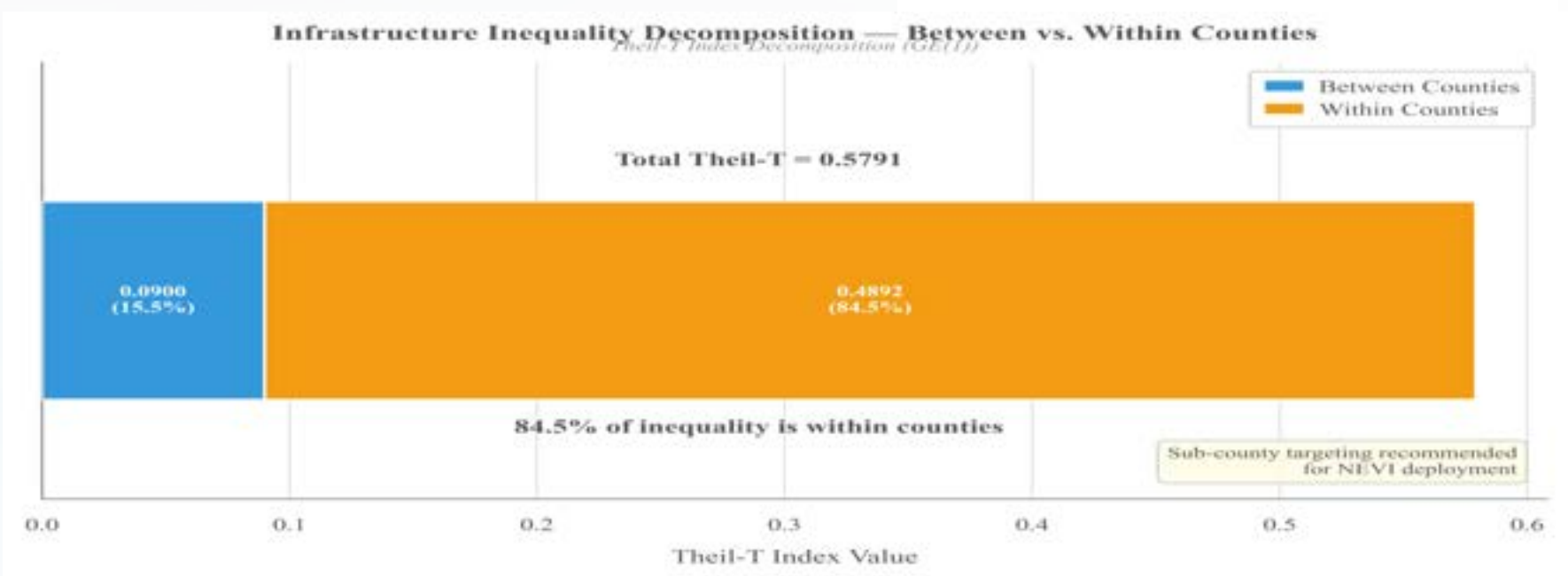
5-lane DQ panel + 4-panelist manuscript advisory — internal methodology review

The Theil Decomposition

The Methodological Contribution

Prior work (Choi, Xu, Jiao, 2025) uses Theil as a *scalar* alongside Gini, Lorenz, and Palma. This study exploits the **additive decomposability of the GE(1) class** (Bourguignon, 1979; Shorrocks, 1980) — the first structural decomposition integrated as a design input to an EV allocation framework.

Theil-L robustness check: 82.5% within-county. Decomposition verified to machine precision.



Theil-T Total (top-10 cohort)	Within-County Share	Between-County Share
0.5791	84.5%	15.5%

NEVI Scoring Framework

NEVI Score = $0.40 \times \text{Equity} + 0.35 \times \text{Utilization} + 0.25 \times \text{Cost-Effectiveness}$

Equity—0.40

Anchored on the Justice40 forty-percent-benefit-flow target; strongest federal mandate signal

Utilization—0.35

Driven by validated BEV forecasts — the strongest and most consistent data in the pipeline

Cost-Effectiveness—0.25

Workplace charging demand; carries the most inherent uncertainty — weighted accordingly

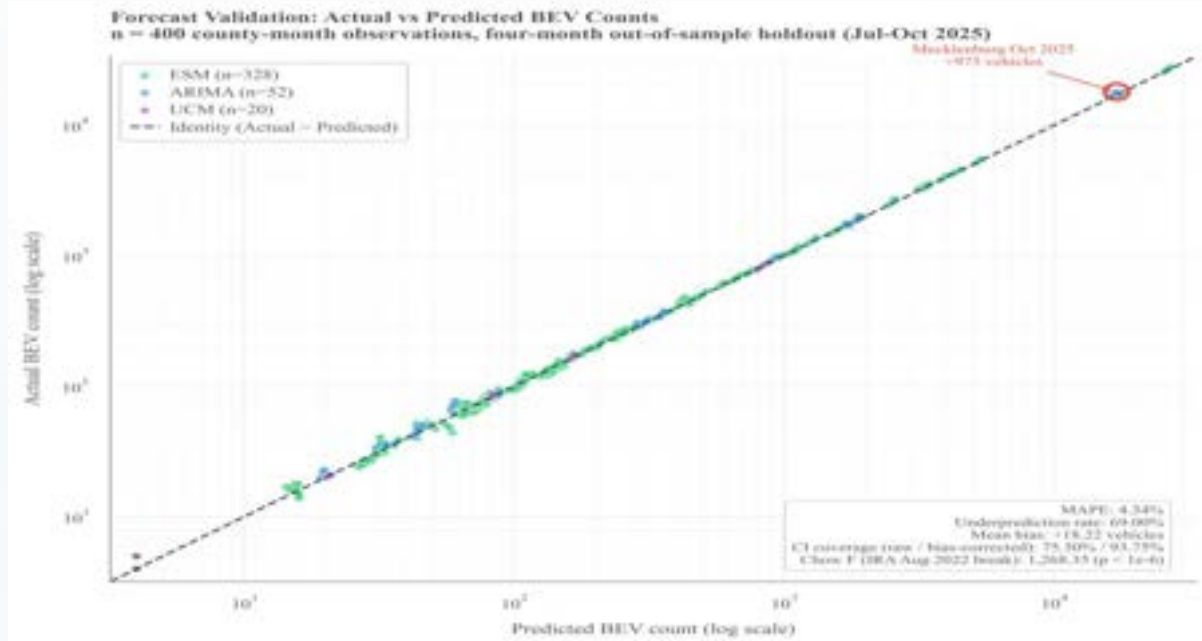
Statistical Independence

VIF max = **1.41** across all three pillars — well below the 5.0 concern threshold. The pillars are statistically independent, not three views of the same underlying factor.



Two-tier output: county ranking + ZIP-level targeting — required by the 84.5% within-county finding.

Forecast Validation: MAPE 4.34% + Chow Break



Key Validation Metrics

95% prediction interval coverage: 75.50% raw → **93.75% after bias correction**. The IRA changed the adoption curve; pre-IRA-trained forecasts are now systematically conservative.

4.34% Out-of-Sample MAPE

n = 400 county-months across all 100 NC counties

69% Underprediction Rate

Mean bias +18.22 vehicles/county-month — a regime signal, not a model defect

Chow Test: F = 1,268.35

$p < 1 \times 10^{-6}$ at August 2022 IRA passage — confirmed structural break

84.5% of EV charging inequality lives *within* counties

Not between them. County-only allocation formulas miss the majority of the equity problem.

Charlotte ZIP 28202

78.64 ports per 10K residents —
downtown core

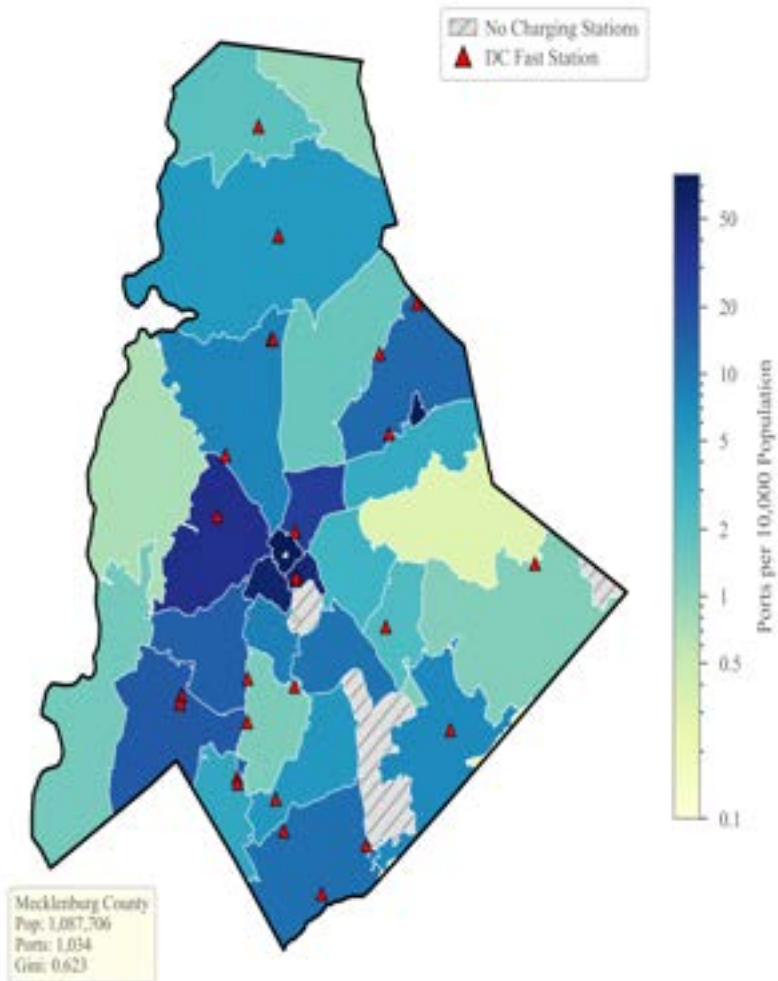
Charlotte ZIP 28215

0.31 ports per 10K residents — 14 miles
away

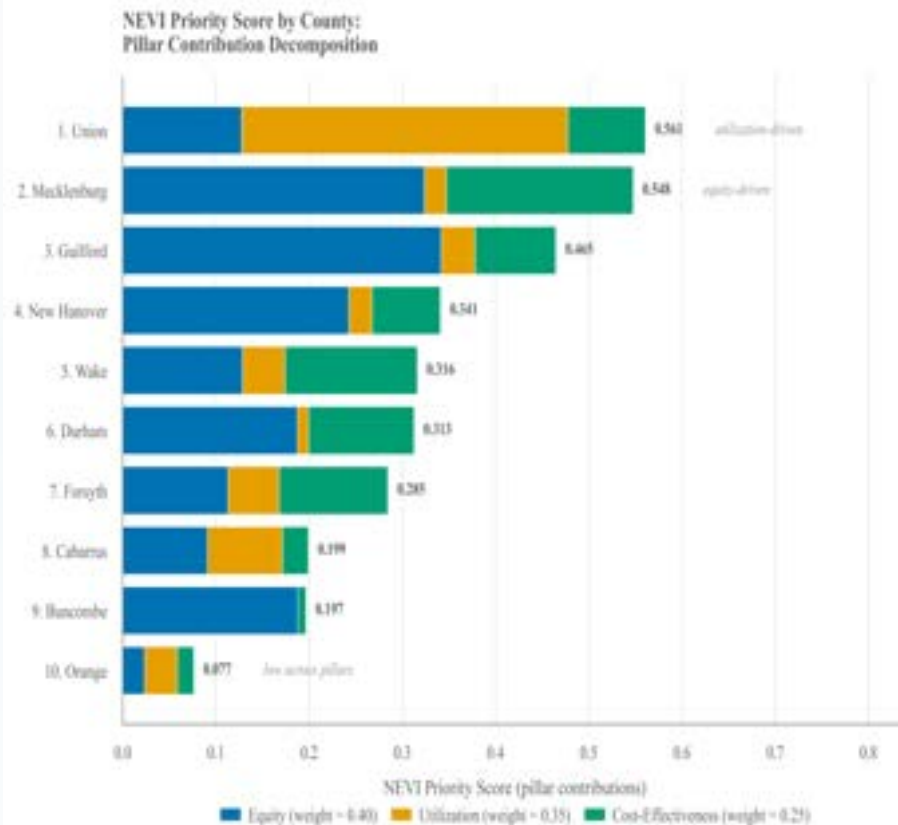
The Gap

250× density difference inside one county — this pattern recurs across the top-10 cohort

EV Charging Port Density — Mecklenburg County



NEVI Priority Score Rankings



Top 3 — Triply Robust

1

Union County

Score: **0.561**

2

Mecklenburg County

Score: **0.548**

3

Guilford County

Score: **0.465**

Also ranked: New Hanover, Wake, Durham, Forsyth, Cabarrus, Buncombe, Orange

✓ Stable across (1) equity-weight sweep 0.30–0.50, (2) cost-effectiveness sub-weight, (3) remote-work multiplier cancellation.

Three County Archetypes

The ranking alone doesn't tell NCDOT *why* each county sits where it does. Three archetypes convert the ranked list into a policy conversation.

● Union — Rank 1

Utilization-Driven

101.5 BEV per port. Needs *more stations*. High demand, insufficient supply everywhere.

📍 Mecklenburg — Rank 2

Equity-Driven

250× within-county density gap. Needs *better-targeted stations* — not just more of them.

○ Orange — Rank 10

Low Across All Pillars

University town, currently adequate. No priority deployment warranted at this time.

📌 The high-equity + high-utilization quadrant is **empty** in NC's top-10 — itself a policy-relevant finding.

Justice40 Equity Overlay

EV Charging Stations on Justice40 Tracts — 10-County Study Area



Equity Geometry — Testable

- **43.0%** of NC tracts designated disadvantaged under CEJST v2.0
- 24.5% of stations in top-10 cohort are in Justice40 tracts vs. **18.5% pop-weighted share**
- Climate sensitivity: 7 of 10 study counties unchanged when CEJST climate category removed
- 23-of-23 federal-standard crosswalk validation suite passed

 CEJST decommissioned Jan 22, 2025 — data preserved via EDGI/PEDP archive. Data lineage is fully traceable.

Five Unique Contributions

Each contribution is substantiated by a specific lane of the methodology review process. Nothing is asserted; everything is verified.

1 First Structural Decomposition of Theil Applied to EV Charging

Additive GE(1) decomposition integrated as a design input to allocation — not merely a scalar inequality measure

2 First HUD USPS / Area-Weighted Crosswalk Integrating CEJST + AFDC at ZCTA Resolution in NC

23/23 federal-standard validation checks passed; data lineage fully traceable

3 NEVI-Specific Three-Pillar Formulation with Documented VIF Orthogonality

VIF max 1.41; triply-robust top-3 rankings confirmed across three independent perturbations

4 Empirical Validation That Within-County Inequality Dominates (84.5%)

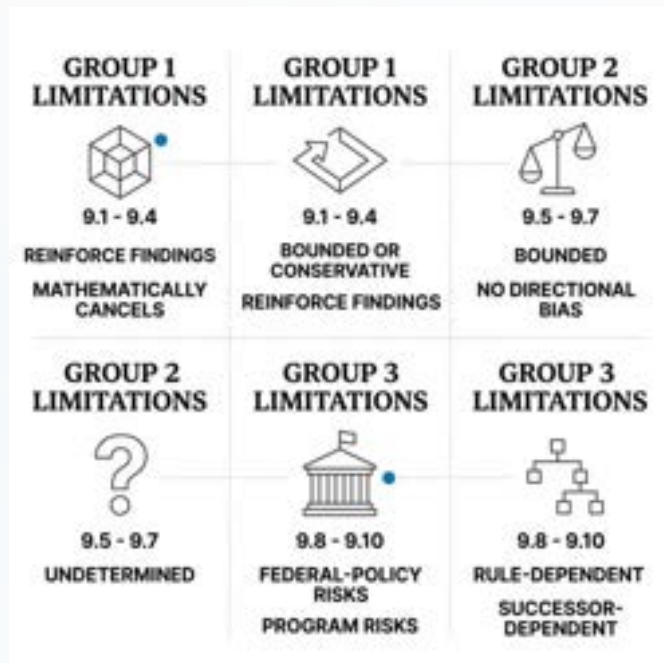
Confirms county-only allocation is structurally insufficient; two-tier targeting is required

5 Climate Sensitivity Analysis Demonstrating CEJST Robustness for NC

7 of 10 study counties show zero rank impact when the most contested CEJST category is removed

Limitations: Direction-of-Bias Framing

A limitation with a direction is a disclosure. A limitation without one is a wound.



10 Limitations • All Named • All Directional

§§9.1–9.4

Reinforce the findings — bounded, cancels mathematically, or conservative. Direction: toward understatement of effect.

§§9.5–9.7

Bounded with no directional bias, or undetermined. No systematic threat to headline rankings.

§§9.8–9.10

Federal-policy or program risks — rule-dependent or successor-dependent. External to the analytical model.

✓ No limitation in this list directly undermines the headline top-3 rankings.

Stakeholder Implications

Six stakeholder groups. One framework. One verification chain. Each group receives a defensible artifact matched to their decision context.



NCDOT / FHWA

Defensible ranking: VIF 1.41, MAPE 4.34%, triply-robust top-3



County Planners

ZIP-level gap analysis for grant applications and site selection decisions



Academic Community

Reproducible pipeline + Claims-to-Evidence Matrix + methodology panels



Justice40 Auditors

23/23 crosswalk validation + CEJST archival pathway fully disclosed



Private Operators

Competitive baseline anchored on Feb 2026 AFDC pull at ZIP resolution

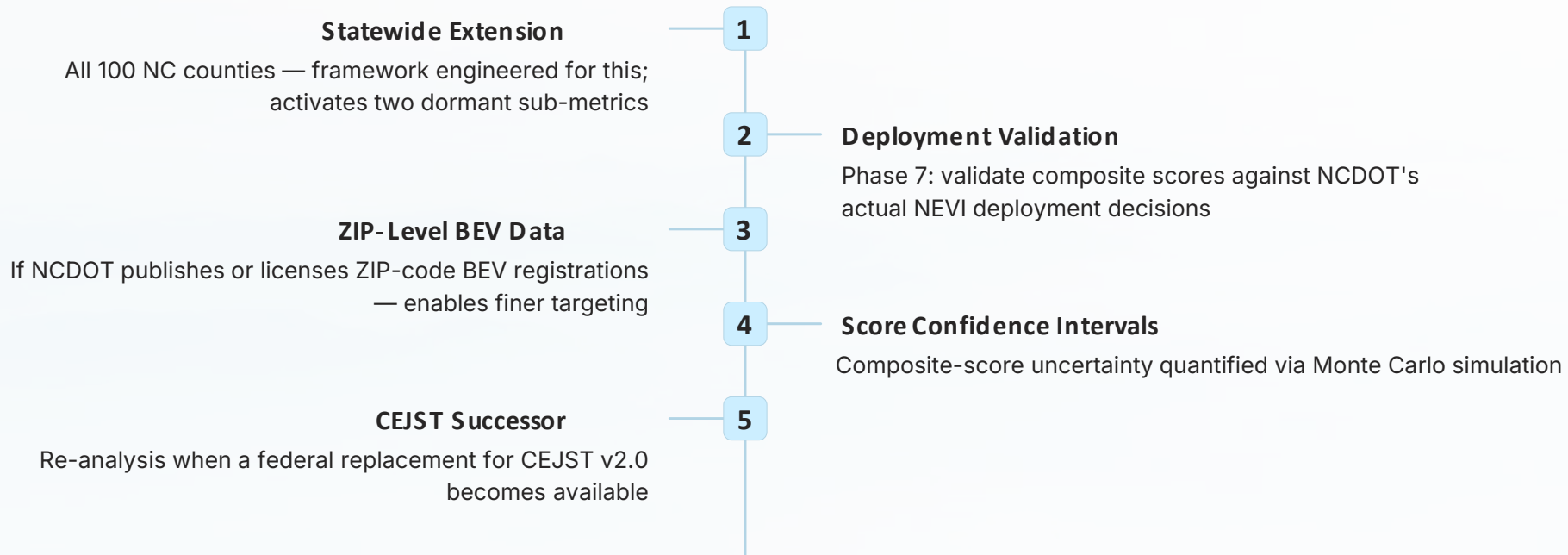


Justice40 Communities

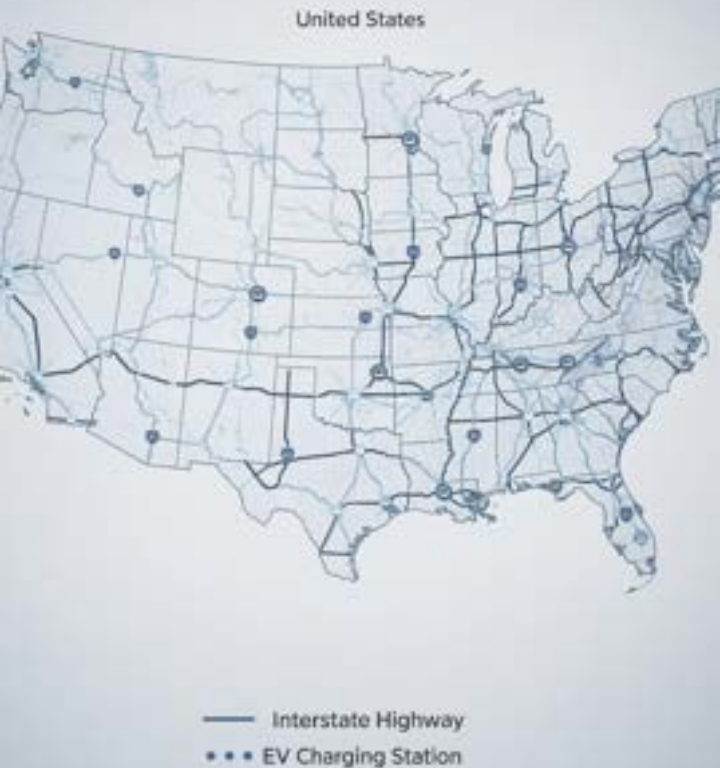
Procedural transparency; fully reproducible from public repo

Future Work: NC-Bounded

The most immediate extensions stay within North Carolina. Each future-work item carries a pointer in the paper's Limitations section — none is required for current findings; each strengthens what's already there.



United States EV Charging Network



ACT 4 — THE NATIONAL PICTURE

Where This Could Go: \$5B · 50 States

The Data Infrastructure Already Exists—Nationally

- **State DOT registration analogs** — every state has one
- **NREL AFDC API** — national coverage
- **Census LEHD LODES** — national origin-destination flows
- **CEJST v2.0 designations** — national; archived via EDGI/PEDP

The Methodology Is the Deliverable

State-by-state calibration requires only tuning the equity weight against each state's Justice40 priorities. The analytical pipeline transfers **without modification**. North Carolina is the proof of concept. The framework's eventual utility is the **full federal NEVI program**.

Project Evaluation: Outcomes & Learning

Deliverable Artifacts

- County-level NEVI Priority Scores for top-10 cohort
- ZIP-level gap analysis for site selection
- Reproducible Python pipeline — 27 scripts, uv lockfile
- Claims-to-Evidence Matrix + two methodology reviews (Appendices A + B)
- Public GitHub repo · 47 publication-grade figures

Student Learning — Four Levels

Analytical

Theil decomposition, Chow test, VIF, area-weighted interpolation

Methodological

Designed AI-assisted role-play internal review processes

Federal Literacy

NEVI, Justice40, CEJST, LODS program fluency

Project Management

Five-phase architecture, phase-gating, scope discipline

What I'd Do Differently

Earlier Literature Audit

The Theil novelty claim required narrowing after a literature check identified Choi, Xu, Jiao (2025) mid-project. An earlier deep audit would have sharpened the contribution framing from the start.

Scope as Constraint, Not Choice

The 10-county analytical window is a constraint of the available data and timeline — not a deliberate design decision. Statewide scope was the target; it remains the primary future-work item.

CLOSING

The Policy That Emerges from the Data

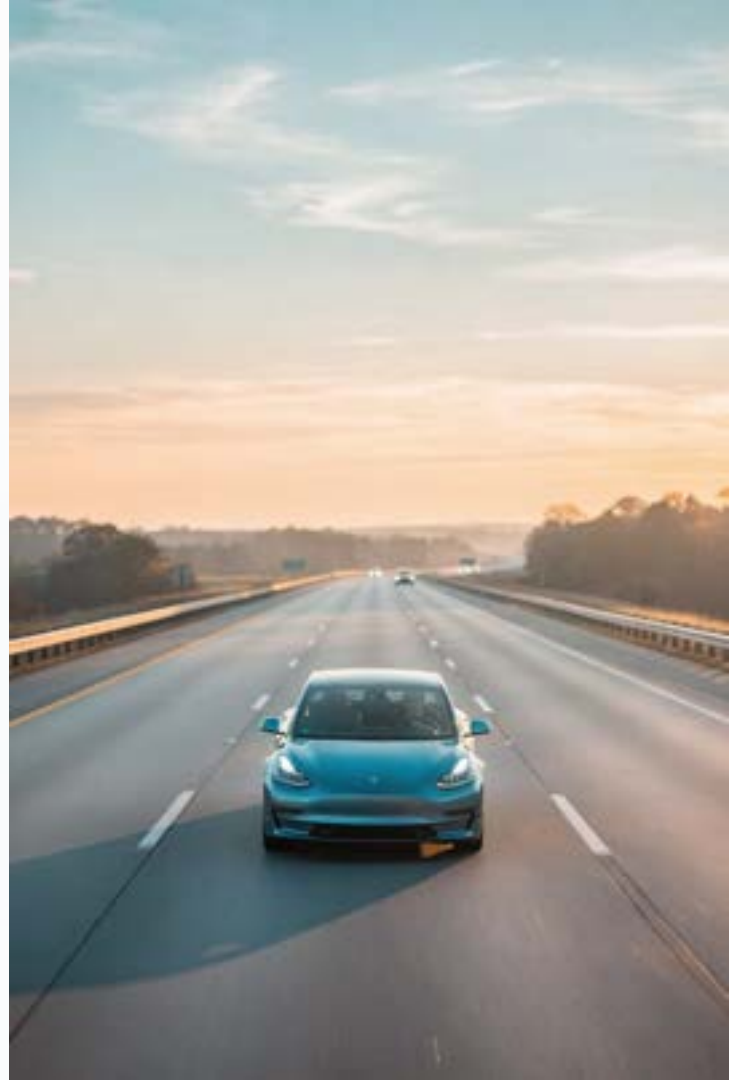
The \$109M NEVI question has a defensible answer for North Carolina — **Union, Mecklenburg, and Guilford** — with the methodological depth to substantiate at the federal-review level.

What the Framework Provides

An **analytical floor** — reproducible, audit-ready, and engineered for extension to all 50 states

What NCDOT Retains

Legitimate non-analytical factors: right-of-way, supply chain, political prioritization — the two are **complementary, not competing**



Acknowledgments & Repository

With Thanks

- **Faculty Advisor:** Dr. Majed Al-Ghandour, Fayetteville State University
- **Data Providers:** NCDOT, NREL, U.S. Census Bureau, CEQ (CEJST archive via EDGI/PEDP)

Public Repository

github.com/wolfieman/ev-pulse-nc

- Polyform Noncommercial 1.0.0 license
- 47 publication-grade figures
- uv lockfile for full reproducibility
- Full paper (~108 pages) available in repo
- AI Methodology: Disclosed in full in Appendix B

Reproduce Any Claim

Every number in this presentation — 4.34% MAPE, VIF 1.41, 84.5% Theil, top-3 Union / Mecklenburg / Guilford — is traceable to a committed script and a processed data file in the public repository.

A federal reviewer can re-execute the entire pipeline from scratch.



Thank you!

github.com/wolfieman/ev-pulse-nc

<https://github.com/wolfieman/ev-pulse-nc/research-paper>

